SMART HOME INTEGRATION IN EXPERIENCE LAB

Reflection

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1. Introduction

This document presents a reflection on my internship at Thomas More Kempen, within the Research Group Mobilab & Care, carried out during the academic year 2025–2026. The project I worked on was titled Smart Home Integration in Experience Lab, and its central goal was to develop a working, demonstrable, and reusable smart home prototype that supports safer night-time routines for older adults living independently at home.

The document is organised into two main parts. The substantive reflection examines what was accomplished during the internship, what value the prototype brings to Mobilab & Care and its future users, what remains to be done, and what advice I would give for the continuation of the project. The personal reflection covers what this experience meant to me personally, the research process, the technical and soft skills I developed, the professional communication workflows I navigated, the human significance of the project, how I grew, and the challenges I faced along the way.

2. Substantive reflection

2.1. What Was Accomplished

The final deliverable of the internship is a fully working local smart home platform built on Home Assistant, running on a Raspberry Pi 5 and deployed in the Experience Lab apartment. The system integrates a Zigbee network of five motion sensors and three manual buttons, smart lights covering the bedroom, hallway, bathroom, kitchen, and living room, a Nuki smart lock, voice control through Amazon Alexa via a Matter Hub bridge, and external notifications through Telegram and WhatsApp.

On top of this infrastructure, twenty-six automations implement the core use cases of the project: night path lighting that guides a resident from the bedroom to the bathroom at low brightness and warm color temperature; a bathroom check-in flow that monitors inactivity and escalates to a full help mode if the resident does not respond; room-based adaptive lighting that adjusts brightness and color temperature depending on whether night mode is active; and a weekday and weekend scheduling system for night mode combined with an adaptive Python-based learning component. That component records activation and deactivation events over a configurable window and generates recommended schedule times, which are presented to the user for acceptance rather than applied automatically.

A four-view Home Assistant dashboard supports daily control, recommendation management, spatial testing, and room-based inspection. A test mode helper compresses timing-sensitive automations from minutes to seconds, making demonstration and validation practical.

2.2. Value for the Organization and Users

For Mobilab & Care, the prototype provides a concrete, demonstrable foundation that can be presented to partners, used in future student projects, and extended in research. Because all logic is locally hosted and built on open-source components, the system does not depend on paid subscriptions or cloud platforms for its core safety functions. The hardware cost of the added components is approximately €680, realistic for a home installation.

For the intended end users, older adults living independently, the prototype addresses a specific and high-relevance situation: the night-time path from bedroom to bathroom. Rather than trying to automate the entire home equally, the system focuses on a small number of meaningful routines where technology can provide genuine safety support without becoming intrusive. The use of gentle, warm lighting during the night, combined with a humane escalation flow that gives the resident multiple ways to cancel a false alarm, reflects a design philosophy centered on dignity and the right to remain autonomous.

2.3. Remaining Work and Future Extensions

The prototype was delivered as a finished and demonstrable system, but several directions remain open. The most immediate is the voice interface: Alexa's Dutch-language mode does not support freely customized spoken responses. A local voice platform, such as Home Assistant Voice, would resolve this and remove the last cloud dependency.

Further research directions I would suggest include context-aware adaptation of the night mode schedule based on calendar or weather data; integration with the Nobi fall-detection device already in the Experience Lab; and extending the system to multi-resident households, where the current single-resident assumption no longer holds. The adaptive learning component is designed to be replaced by a more sophisticated model without changing the rest of the system.

2.4. Advice for the Principal

My main advice for Mobilab & Care is to treat reproducibility as a first-class concern from the start of any continuation. The system I delivered is documented in a realization document, an installation and maintenance manual, a full system setup and configuration guide, because I learned that institutional knowledge disappears when a student leaves. Future teams should update these documents as the system evolves.

I also recommend arranging network permissions, particularly static IP reservations for the Raspberry Pi, the Nuki Bridge, and the Matter Hub host, before technical work begins. In my case, network-related delays from the IT department accounted for a significant portion of the timeline. Anticipating this dependency early would meaningfully reduce friction for any successor project.

3. Personal reflection

3.1. What the Internship Meant to Me

My Bachelor's degree in Applied Computer Science includes IoT as part of the curriculum, and I had taken two dedicated IoT courses. But more than that, IoT had become a genuine personal interest, something I researched and experimented with on my own, beyond what the classes required. By the time I started this internship, I already had working knowledge of MQTT, understood how message brokers and publish-subscribe patterns work, and was familiar with home automation concepts at a practical level. IoT was not unfamiliar territory for me; it was territory I had chosen to explore because I enjoyed it.

What was outside my scope was KNX. In week one, I presented ideas built on the existing KNX infrastructure in the Experience Lab, a professional-grade wired building automation standard that requires specialist hardware and configuration. My mentor redirected me toward a low-end, wireless, portable solution that could work in any home without depending on existing infrastructure. Looking back, this was exactly the right call: KNX is a specialist ecosystem, and the project brief was about accessible, replicable technology. But accepting the change in week one, after already investing time in the original direction, was the first real professional test of the internship.

My architecture background contributed something less obvious but genuinely useful throughout the project. In architecture, design is inseparable from how people actually use a space, how they move through it, what feels natural at two in the morning when you are half asleep, where you expect a light to come on. That user-oriented spatial thinking shaped decisions that would not appear in a technical specification: the choice to route the night path lighting through the hallway rather than just the destination room, the decision to keep bedroom lighting passive rather than motion-triggered, the reasoning behind timer durations long enough for a disoriented person to respond. Architects think about people in buildings. That habit transferred directly to thinking about an older adult moving through a dark apartment at night.

The electrical plan I produced for the Experience Lab also brought an unexpected learning moment. I have drawn electrical schematics before, but using the symbols I was taught during my architecture studies in my home country. Early in the project, I discovered that those symbols do not match the Belgian AREI standard. I had to relearn the notation from scratch and produce the plan in the correct format. It was a small thing in the wider context, but it was a concrete reminder that professional conventions are local, and that competence in one context does not transfer without adjustment.

The larger risk of this internship, though, was one my school supervisor pointed out before I even started: my specialization within Applied Computer Science is Artificial Intelligence, and I had chosen an IoT internship. That is not the expected path. There was a genuine concern about whether it was the right fit, whether I was stepping too far outside the trajectory I had been building. I understood the concern, but I took the risk deliberately. I had always wanted to go deeper into IoT than the courses allowed, and I knew that staying only within AI would have been the safer but smaller choice. By the end of the thirteen weeks, I had built a working prototype, written documentation that will outlast the internship, and developed a fluency in a technical domain I care about. The risk was worth it.

3.2. Research as a Core Competency

One of the things that defined this internship from the very first week was the emphasis my mentors placed on research. They consistently pushed me to go deeper: not to find a solution and move on, but to understand why it was the right solution, what the alternatives were, and what the evidence said. That approach shaped everything from the platform comparison in the analysis chapter of the realization document to the voice interface selection and the lighting strategy.

In practice, this meant that before I could implement anything, I had to build a justified case for it. The weighted decision matrices I used to compare Home Assistant against OpenHAB and proprietary ecosystems, or Alexa against Google Assistant, Siri, and Home Assistant Voice, were not exercises in paperwork, they were the result of genuinely exploring each option, testing where possible, and building an argument I could defend. My mentors did not want to hear what I had chosen; they wanted to understand how I had chosen it.

This approach to research, going in with intention rather than with a conclusion already in mind, is something I carry forward as a professional habit. It made the final system more defensible, more coherent, and more honest about its own limitations.

The research phase was also where the academic literature became directly useful. Reading about ageing in place, fall prevention, and the psychology of technology acceptance for older adults was not background reading, it directly shaped design decisions. For example, the finding that gentle guiding light reduces fear of falling at night (Thölking et al., 2020) was the specific reason the system uses 15–25% brightness at warm color temperatures during the night path, rather than simply switching a light on. That kind of connection between evidence and implementation is what my mentors were asking for, and learning to make it deliberately has been one of the most lasting things I take from this experience.

3.3. Technical Skills Developed

This internship gave me hands-on proficiency in a set of technologies I had not worked with before. The table below summarises the main areas:

Home Assistant	Platform configuration, entity management, helper states, YAML automations and scripts, dashboard design across four distinct views
MQTT & Zigbee	Mosquitto broker setup, Zigbee2MQTT integration, multi-vendor device pairing and troubleshooting, event tracing through MQTT topics
Python / AppDaemon	Event-driven scripting, Home Assistant API queries, robust-statistic analysis for night mode learning, datetime helper updates
Matter Hub	Exposing Home Assistant entities as Matter devices, Alexa discovery and routine binding, local-network Matter bridge operation

System administration	Raspberry Pi OS setup, microSD imaging and restore, static IP coordination, SSH access, full system migration from Pi 3B to Pi 5

Electrical planning	Adaptation from Latin American architectural notation to Belgian AREI standards for the Experience Lab control plan

Beyond the individual tools, what I developed was the ability to work across layers: from the physical placement of a sensor, through the communication protocol that carries its events, to the automation logic that acts on them, to the dashboard that makes the state visible. Understanding the full stack, not just one piece of it, is the skill I value most from the technical side of this internship.

3.4. Soft Skills Developed

The internship addressed a range of professional competencies that go well beyond technical content. The ones I developed most concretely are described below.

Adaptability	Redirected in week 1, hardware pivot in week 5, Alexa integration pivot in week 6, each required absorbing a change and reorienting without losing momentum

Written communication	Produced a realization document, installation and maintenance manual, full system setup guide, and user manual, each for a different audience and purpose

Structured decision-making	Weighted comparison matrices, justified technology choices, documented trade-offs, research translated into defensible decisions

Presentation and demo	Weekly sync meetings, formal project plan presentation, and scenario-based live demonstrations of the prototype to mentors

Time management	Balancing implementation, documentation, and testing across 13 weeks with a shifting set of constraints and dependencies

3.5. Professional Communication and Interdepartmental Workflow

Working inside an institutional environment, a university research group on a company campus, meant that many things I needed could not simply be done independently. They required coordination with other departments, formal requests, and follow-up. Learning how to navigate this was one of the less expected but genuinely valuable parts of the internship.

The most frequent point of contact was the IT department. Every device I needed to connect to the network, both Raspberry Pi devices, the Amazon Echo Spot, the Nuki Bridge, had to be registered through IT. This meant submitting MAC addresses, waiting for network reservations to be applied, and sometimes re-testing after a network change invalidated a previous configuration. The static IP reservation for the Nuki Bridge, in particular, was critical: without it, the Home Assistant integration would fail every time the network restarted and the Bridge received a new address.

What I learned from this is that in a professional context, the technical solution is rarely the only thing that needs to be managed. Knowing when to wait, how to frame a request clearly, how to follow up without being disruptive, and how to keep progress moving on other fronts while a dependency resolves, these are skills that do not appear in any curriculum but matter enormously in practice.

Communication with my mentors followed a structured rhythm. Weekly sync meetings served as the primary checkpoint: I would present the current state of the system, raise any blockers, and get direction on the next steps. What I appreciated about my mentors was their combination of high expectations and genuine support. They pushed me to go deeper in the analysis, to justify choices rather than just make them, and to think about the system's reusability beyond the internship period. At the same time, when I ran into genuine blockers, hardware limitations, network access issues, integration challenges, they were responsive and decisive. The formal approval processes, such as the hardware upgrade proposal I submitted for the Raspberry Pi 5, taught me how to write a clear, justified request for something I needed rather than simply asking informally.

I also learned that communication with different departments requires different registers. A request to IT needs to be concrete and technical: which device, which MAC address, which network, which port. A progress update to a mentor needs to be contextual and honest: what works, what does not, what the dependency is, and what the plan is. Writing for a future user in a manual requires a different voice again patient, instructive, and free of assumed knowledge. Managing those shifts in register across the same week was a form of professional communication training I had not experienced in an academic context.

3.6. The Human Impact: Why This Project Matters

Throughout this internship, I kept returning to the same question: why does this actually matter? And the answer, the real answer, not the project brief version, is something I want to reflect on honestly, because it is what made the difficult weeks worth pushing through.

The system I built is designed for older adults who want to stay in their own homes as long as possible. That desire to remain in a familiar space, surrounded by personal history, maintaining independence and routine is one of the most deeply human things there is. Research on ageing in place is clear: people who remain in their own homes tend to have better psychological outcomes, stronger senses of identity, and greater feelings of autonomy than those who move into care facilities earlier than necessary. Technology that supports safe independent living at home is not just convenient, it can genuinely extend the period during which a person can remain who they are, in the place that matters to them.

But this project is not only about the person living there. It is also about the family members who worry every night. A parent who lives alone, an elderly spouse who gets up at three in the morning, the anxiety that surrounds those situations is real and persistent. A system that can send a Telegram message to a family member when help mode activates, or that unlocks the front door automatically so a responder can enter without delay, changes what it means to be a worried family member. It gives them something concrete: a safety net, not perfect, but real.

The night-time bathroom trip is a small thing. It takes a few minutes, happens in the dark, and is forgotten by morning. But for an older adult with reduced balance or a fear of falling, it is a moment of genuine vulnerability. The design decision to use gentle, warm, low-level lighting rather than harsh full-brightness activation is not just an aesthetic choice, it is grounded in research showing that guiding light reduces fear of falling and improves sleep-related outcomes. The system does not treat the resident as a problem to be monitored. It treats them as a person who deserves to feel safe, calm, and in control in their own home.

Working on a project with that kind of purpose changed how I approached every design decision. When I was choosing timer durations for the bathroom check-in flow, I was not thinking about a system requirement, I was thinking about a real person who might be disoriented at three in the morning, who should have enough time to respond without feeling surveilled. When I designed the help escalation to give the resident multiple ways to cancel a false alarm, motion, hallway movement, a voice command, a button, I was thinking about what it would feel like to be that person. That human orientation, sustained across thirteen weeks of technical work, is one of the things I am most proud of.

Ageing in place is not a technical problem with a technical solution. It is a human situation that technology can support, imperfectly but meaningfully. I believe the prototype I built contributes something real to that space, not because it is perfect, but because it is honest, grounded in evidence, and designed with the person at the center rather than the technology.

3.7. How I Grew

The most significant growth I experienced during this internship was learning to work steadily through uncertainty. For several weeks, particularly weeks seven and eight, I could not progress on the core implementation because the lab network was inaccessible. I could have let that derail me. Instead, I used the time to advance the documentation, which in the end turned out to be one of the strongest parts of the project. That shift in mindset from seeing blocked time as lost time to seeing it as redirected time is something I did not have at the start of the internship.

I also grew in my relationship with hardware. When the Raspberry Pi 3B reached its memory ceiling under the combined load of Zigbee2MQTT, Matter Hub, AppDaemon, and an increasing number of automations, I diagnosed the problem, proposed the hardware upgrade to my mentors, wrote the formal justification, and executed the migration, restoring from backup on the new device, without losing any configuration. That end-to-end ownership of a hardware problem gave me a confidence with embedded systems and system administration I did not have before.

On a more personal level, I grew in my ability to hold two things at once: the technical detail and the human purpose. It is easy, when debugging a serial port path or tracing an MQTT topic, to forget what the system is actually for. The research that my mentors pushed me to do, grounding design choices in evidence about older adults, fear of falling, and the meaning of home, kept that connection alive throughout the project.

3.8. Problems I Faced and How I Addressed Them

The most persistent challenge was the network environment. Working inside a campus network meant that device registration, static IP reservations, and firewall rules required IT department coordination. This created a recurring pattern where I was technically ready but blocked by an external dependency. I addressed this by always keeping a secondary track of work available when I could not test in the lab, I was writing documentation or testing components on a home network. By the time network access was restored in week nine, the documentation scaffolding was far enough along that the final implementation sprint was significantly more efficient.

The Alexa integration required a full pivot midway through the project. The first approach I implemented required a subscription service, which my mentors asked me to replace. Finding and validating an alternative, the Home-Assistant-Matter-Hub bridge, took several weeks of research and testing across different network environments. What helped most was reframing the constraint as a design requirement: if the system had to work without a subscription, then the solution was also more sustainable and privacy-preserving, which is a genuine improvement.

One challenge that surprised me was the electrical diagram. I had professional experience drawing electrical plans from my architecture work, but I discovered early in the internship that the symbols I had been trained to use in my home country do not match the Belgian AREI standard. I had to relearn the notation from scratch. It was a small thing in the context of the whole project, but it reminded me that professional standards are local, and that expertise in one context does not always transfer directly to another.

3.9. General Conclusion

This internship was, in the most honest sense, the experience I had been building toward without fully knowing it. The IoT curiosity that started in two university courses and grew into a personal hobby found a real application here, not as a side project, but as a professional deliverable with a genuine purpose. The architecture background that I had never expected to use in a computer science context turned out to shape the most important design decisions of the project. And the risk of choosing an IoT internship when my specialization is AI, which worried my school supervisor and gave me pause too, turned out to be exactly the kind of productive discomfort that forces real growth.

What I take away from these thirteen weeks is not a checklist of technologies I can now list on a CV, though that is part of it. It is a different kind of confidence, the kind that comes from having navigated uncertainty without losing direction. From having been redirected in week one and finding a way to redirect my energy with it. From having spent weeks seven and eight without lab access and choosing to use that time well rather than wait it out. From having diagnosed a hardware failure, proposed a solution, written the formal justification, executed the migration, and kept the project moving. From having written documentation that I genuinely believe the next person will find useful, rather than documentation that fulfils a requirement.

The research depth my mentors demanded from me was demanding precisely because it mattered. Every weighted comparison, every literature reference, every justified design decision made the final system more coherent and more defensible. That habit of researching with intention rather than reaching for the first available answer is the professional practice I am most glad to have developed here.

And underneath all of it, the technical work, the documentation, the network issues, the hardware migrations, was a project that genuinely matters to real people. Older adults who want to stay in their own homes. Family members who worry every night. The experience of holding that human purpose in mind, consistently, across thirteen weeks of technical work, taught me something that no course has taught me: that the most meaningful work is the kind where you never lose sight of who it is for.

Looking back at thirteen weeks of this internship, what I am most proud of is not the number of automations I wrote or the technical complexity of the final prototype. It is the fact that I delivered a system that is honest about what it can and cannot do, documented in a way that gives the next person a real starting point, and grounded in a genuine understanding of why it matters. Behind every motion sensor and every lighting automation is a real person who deserves to feel safe, independent, and at home and that purpose is what I will carry forward from this experience.
